INCH-POUND

MIL-DTL-12316F
28 May 2026
SUPERSEDING
MIL-C-12316E
10 August 1984

MILITARY SPECIFICATION

CASE, MAP AND PHOTOGRAPH

This document is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers a cotton duck map case and photograph with shoulder strap for sketching.

1.2 Classification. The map case is of one (1) type and one (1) size.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be sent to trpsptctstandardiz@dla.mil. Since contact information can change, verify the currency of the address information using ASSIST https://assist.dla.mil.

AMSC N/A FSC 8460

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COMMERCIAL ITEM DESCRIPTION

A-A-3174 - Plastic Sheet, Polyolefin
A-A-55301 - Webbing, Textile, Textured or Multifilament Nylon
A-A-59826 - Thread, Nylon

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-543 - Buckles, Tongueless and Web Strap
MIL-DTL-3390 - Rings, Dee
MIL-DTL-5038 - Tape, Textile and Webbing, Textile, Reinforcing, Nylon
MIL-DTL-l0884 - Fasteners, Snap
MIL-PRF-17337 - Webbing, Textile, Woven Nylon
MIL-DTL-32439 - Cloth, Duck, Textured Nylon
MIL-DTL-32075 - Label: For Clothing, Equipage, and Tentage, (General Use)

(Copies of these documents are available online at https://quicksearch.dla.mil)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

U.S. ARMY, DEVCOM – SOLDIER CENTER

2-2-220 - Case, Map and Photograph, Assembly, Illustration and Details
2-2-221 - Case, Map and Photograph, Details and Sections
2-2-222 - Case, Map and Photograph, Map and Photograph Envelopes

(Copies of drawings are available from the U.S. Army, DEVCOM - Soldier Center, ATTN: FCDD-SCD-PEC, 10 General Greene Avenue, Natick, MA 01760-5019.)

(Comments, suggestions, or questions on this document should be addressed to: U.S. Army, DEVCOM Soldier Center, ATTN: FCDD-SCD-PEC, 10 General Greene Avenue, Natick, MA 01760-5019.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION

AIA NASM 43770/1- Snap Hooks, Wire Body, Fixed Loop Eye, Flat Spring Closure, with Retainer

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(Copies of these documents are available online from https://www.aia-aerospace.org)

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC TM20A - Fiber Analysis: Quantitative

(Copies of this document is available online at https://www.aatcc.org)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org)

ASTM INTERNATIONAL

ASTM D1777 - Standard Test Method for Thickness of Textile Materials
ASTM D3990 - Standard Terminology Relating to Fabric Defects
ASTM D6193 - Standard Practice for Stitches and Seams

(Copies of these documents are available online at https://www.astm.org)

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

T-411 OM -Thickness (caliper) of Paper, Paperboard, and Combined Board

(Copies of this document are available online at https://www.ansi.org)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Standard sample. Unless otherwise indicated, the finished map case shall match the standard sample for shade and appearance, and shall, be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum

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extent possible, provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

3.4 Design. The map case shall be manufactured in accordance with Drawings 2-2-220, 2-2-221 and 2-2-222.

3.3 Basic and secondary material.

3.2.1 Basic material. The cloth for the map case shall be nylon duck, plain weave conforming to Type I, Class 3, Style H in accordance with MIL-DTL-32439.

3.2.2 Webbing, textile, nylon. The webbing shall be Tan 499 (see 6.2) and shall conform to the following:

2-inch width (for shoulder pad) in accordance with Class 2 of MIL-PRF-17337
1-inch width (for shoulder strap and for dee ring chape) in accordance with Type III of A-A-55301.

3.2.3 Tape, textile, nylon. The tape for the keeper loops and binding shall be Tan 499 (see 6.2) and shall conform to Type III, Class 2, 1-inch width in accordance with MIL-DTL-5038.

3.3 Components.

3.3.1 Thread, nylon. The thread for all stitching shall conform to A-A-59826, Type I or Type II, Class B, Size E for bartack and binding only, and Size F for all other operations. The color shall be Tan 499.

3.3.2 Fasteners, snap. The snap fasteners shall conform to MIL-DTL-10884, Style 2.a and Style 2A,b, Styles 1B and 2a., Finish 2, any construction.

3.3.3 Buckles, tongueless, 1-bar. The buckles shall conform to Type I, Style 2, Class 1, Size l-inch of MIL-DTL-543.

3.3.4 Rings, loop. The loop rings shall be in accordance with Waterbury P/N 000-12-20-21-883 or equal and shall have a black enamel finish.

3.3.5 Snap hook (2-1/2 by 1/4-inch). The snap hook shall be in accordance with NASM 43770/1 and have a black enamel finish.

3.3.6 Rings, dee, (1-inch by 3/4-lnch). The dee rings shall conform to Class 1, Configuration K of MIL-DTL-3390.

3.3.7 Envelopes. Each map case shall be provided with envelopes in the quantity and sizes as specified herein.

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3.3.7.1 Map envelopes. Three (3) map envelopes, 20-1/2 inch by 23-inch shall be fabricated from polyethylene sheets or tubing conforming to Type II, Class 1, Grade C, Finish 1, 0.0060-inch nominal thickness, conforming to A-A-3174. The envelopes shall be of such transparency that the contents can be easily read or interpreted without being removed.

3.3.7.2 Photograph envelopes. Nine (9) opaque photograph envelopes, 10-1/2 inch by 11-3/4 inch, shall be fabricated from 0.01-inch to 0.002-inch thick sheets of cellulose acetate or cellulose acetate butyrate containing 2 to 3 percent carbon black. The envelopes shall be free of breaks or cracks and shall show no signs of tackiness when tested for cold and heat resistance as specified in 4.6.1.

3.4 Marking.

3.4.1 US and identification markings. The map case shall have the letters "US" printed in black (approximately 1-inch in height) shall conform to Type IV, Class 9 of MIL-DTL-32075 in the location as indicated on Drawing 2-2-220. The identification markings shall also be in the location as indicated on Drawing 2-2-221 conforming to Type IV, Class 5 of MIL-DTL-32075 and contain the information below. As an alternate, Type III (heat transfer) markings may be used.

CASE, MAP and PHOTOGRAPH OCP
Contract Number:
National Stock Number (NSN):
Contractor's name:
Lot number:

3.4.2 Bar code hang tag. Each map case shall be individually bar coded with a Type VIII, Class 17, of MIL-DTL-32075. The bar coding element shall be a 13-digit National Stock Number (SNS). There shall be a 2-digit Universal Product Code (UPC) number assigned for all NSNs by the contracting activity. The bar code type for NSN and UPC shall be a medium to high code density, shall be human-readable interpretation (HRI), and shall be located so that it is completely visible on the item when it is folded and packaged causing no damage to the map case.

3.4 Construction. The map case shall conform to the drawings listed in paragraph 3.4 and as specified herein.

3.4.1 Stitches, seams and stitching. All stitching shall be as specified on drawing 2-2-221 in conformance to Type 301 of ASTM D6193 and shall be 6 to 8 stitches per inch. The stitching and seams shall be consistent, exhibiting a uniform appearance and seam allowances shall be maintained with seams sewn so that no raw edges, runoffs, please or open seams occur. Thread tensions shall be maintained so there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching. All seams shall start and finish evenly so no puckering of the materials sewn should occur. All thread ends shall be trimmed to 1/4-inch maximum in length.

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3.4.1.1 Thread breaks. All thread breaks in stitching shall be overstitched not less than 1-inch at each break.

3.4.1.2 Stitching ends. All stitching at the ends shall be overstitched not less than 1-inch except where ends are turned under in a hem, held down by other stitching or where stitching is performed automatically. Where stitch patterns such as box, or box with cross-stitch are performed automatically, at least three (3) or more tying, overlapping or backstitches shall be used to secure the ends.

3.4.1.3 Lubrication of thread. The addition of any lubricant to the polyester sewing thread prior to or during the sewing operations is prohibited.

3.4.2 Bartacks. To maintain durability and functionality, bartacks shall be correctly placed to reinforce points of strain as specified in Table I and on drawings. Bartacks shall be 3/4-inch wide. Nominal lengths and stitches per bartack shall also be as specified in Table I. Bartacks shall have a minimum of three (3) rows of stitching and shall be free from thread breaks and loose or tight stitching. All webbing that is inserted into seam allowance shall be bartacked the full width of the webbing.

TABLE I. Bartack placement.

Bartack location	Size of bartack (inches)	Stitches per bartack (min.)	Number of bartacks	Total number
Left and right top edge of side and front panel	3/4	36	2	4
Left and right dee ring chape on back panel	1-1/2	68	1	2
Folded attachment of snap hook loops at each end of shoulder strap	3/4	36	1	2

3.4.3 Setting of metallic snap fasteners. A hole shall be pre-punched through the material before inserting the stud eyelet tube or the socket eyelet tube. The hole shall be smaller than the outside diameter of the applicable eyelet tube so that the tube must be forced through the hole. pre-punched holes shall not be accomplished by using the stud or eyelet tubes. Fastener components shall be securely clinched without cutting the material. The roll of the clinched eyelet of the assembled fastener component shall not possess more than two (2) scores (Splits).

3.5 Finished dimensions. The finished map case shall conform to the measurement as specified on the drawings (see 3.4).

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3.6 Workmanship. The finished map case shall be uniform in quality and shall conform to the quality of product established by this specification. The occurrence of defects as specified in 4.4 and 4.5, shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 Inspection conditions. Unless otherwise specified in this specification or applicable procurements documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.3 Types of inspections.

4.3.1 First article inspection. The first article, submitted in accordance with 3.1 shall be inspected for compliance with design, dimensional requirements, examination for defects listed in 4.4 and tested for the characteristics as specified in 4.5 through 4.6.1. If first article is required for the cloth, it shall be in accordance with 4.5, shall be inspected, examined for appearance, color, and finished defects as specified in Table I and tested for the characteristics as specified in Table II.

4.3.2 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5 through 4.6.1 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract and/or order, except where otherwise indicated (see 6.2).

4.4 Visual examination. The finished map case shall be examined for the defects listed in Table I. The lot size shall be expressed in units of map cases. The sample unit shall be one (1) finished map case. Fabric defects are defined in ASTM D3990.

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TABLE I. End item visual defects.

Examine	Defect	Classification	
		Major	Minor
Materials			
Fabric	- Any smash, weak place, broken or missing yarns, multiple floats or open places - Shade bar, or fine or coarse filling bar - Hole, cut, or tear	101 102	 201
Material components			
Webbing and Tape	- Frayed or scalloped edge, holes, cuts, tears, and missing yarns	103	
Hardware			
Snap fasteners	- Not securely clinched so that they may become disengaged from body, or clinched excessively tight cutting material - Fasteners fall to snap in a closed position or to open freely - No more than two scores (splits) removed per snap fastener	104 105	 202
Snap, loop eye	- Spring loose or defective, i.e. causing spring to jam or remain in an open position - Spring does not exert pressure on body at opening - Enamel finish shows orange peels, wrinkles, drops, streaks, or areas of no film, when applicable	106 107	 203
Stitches, seams and stitching			
	Open seams: - 1/2-inch or less - more than 1/2-inch but not more than 1-inch - more than 1-inch NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken and not over-stitched, when there are two or more continuous skipped stitches, or when run-offs occur. On double stitched seams, a seam shall be classified as open when one or both sides of a seam are broken - Run-offs - Thread breaks overstitched less than 1-inch - Any raw edge; mends or darns; needle chews - Wrong seam or stitch type	 108 109 110 111	 204 205 206

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TABLE I. End item visual defects. – Continued

Examine	Defect	Classification	
		Major	Minor
Stitches, seams and stitching (cont'd)			
	- Loose, resulting in a loosely exposed top or bobbin thread		207
	- Tight, resulting in excessive tightness or puckering of fabric		208
	- One stitch less than minimum specified		209
	- Two or more stitches less than minimum specified		210
	– One or two stitches in excess of maximum specified		211
	- Three or more stitches in excess of maximum specified		212
	NOTE: Variation ln the number of stitches per inch, caused by the operator speeding up the machine and pulling the fabric in order to sew over heavy places, heavy seams or in turning corners, shall be classified as minor		
	- Not as specified		213
	- Overstitched less than l-inch (except where ends are held down by other stitching, turned under in a hem or where stitching is performed automatically)		214
	- Secured with less than three (3) tying, overlapping, or back stitches where automatic stitching is performed		215
	Box stitching:		
	- Incomplete:		
	-one end row of stitching omitted		
	-two end rows of stitching omitted	112	
	-any other row of stitching omitted	113	
	Bartacks:		
	- Not running in correct direction		216
Components			
Envelopes	- Any hole, cut, crack, or tear	114	
	- Folding margin not bonded	115	
	- Alternate method, heat seal open, more than 1-inch	116	
	- Two or more envelopes omitted	117	
	- One envelope omitted		217
	- Heat seal improperly applied		218

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TABLE I. End item visual defects. – Continued

Examine	Defect	Classification	
		Major	Minor
Construction and workmanship			
Components, hardware and assembly	- Any required component or operation omitted (unless otherwise specified herein) - Broken or malformed; protective finish omitted; corroded area - Burrs or sharp edges which may damage the item or cause injury in handling - Any required operation not as specified	118 119 120 121	
Binding	- Improperly or loosely applied; ends not finished as specified		219
Pocket compartments	-Partition stitching forming pockets omitted, or obstructing passage	122	
Body flap	- Corners not rounded, or irregularly shaped		220
Shoulder pad	- Not one continuous piece forming double ply - Ends not abutted by more than 1/8-inch		221 222
Markings	- Omitted, incorrect, illegible, misplaced, or character size not as specified		223
Cleanness	- Grease or oi1 stains; thread ends not trimmed throughout		224

4.4.1 Dimensional examination. The map cases shall be examined for conformance to the dimensions specified in the drawings (see 3.4). The lot size shall be expressed in units of map cases.

4.5 End item testing. The map case and envelopes shall be tested for the characteristics listed in Table II. The methods of testing as specified wherever applicable and as listed in Table II shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be one (1) envelope for all physical and chemical tests. The lot shall be unacceptable if one or more sample units or the lot average fail to meet any requirement specified. The sample size shall be in accordance with the following:

Lot size (envelopes)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE II. End item tests.

Characteristics	Requirement Reference	Test method	Number determinations per sample unit
Map Envelope			
Thickness	3.4.1.1	ASTM D1777	--
Photograph Envelope			
Thickness 1/	3.4.1.2	TAPPI T-411	1
Fiber identification	3.4.1.2	AATCC TM20A (see 6.3)	1
Carbon black content	3.4.1.2	AATCC TM20A (see 6.3)	1
Cold and heat resistance	3.4.1.2	4.6.1	1

1/ Measured to the nearest 0.001-inch.

4.6 Methods of inspection.

4.6.1 Cold and heat resistance test. Condition the photograph envelopes at a temperature of -100°F for 1 hour. After conditioning, bend the envelopes on a surface having a 2-inch radius and examine to determine conformance to 3.4.1.2. Then condition the envelopes at a temperature of 125°F for 1 hour and examine to determine conformance with 3.4.1.2. Any break, crack, or tackiness shall be considered a test failure.

4.6.2 Visual shade matching. The color and appearance of the finished map case shall match the standard sample when viewed using the AATCC EP 9, Option C, (see 6.6) with a primary light source simulating the spectral quality of average daylight, CIE Illuminant D65, with a color temperature of 6500 (± 200) Kelvin (K) illumination of 100 (± 20) foot candles. Alternatively, the color and appearance of the map case shall match the standard sample with a primary light source simulating artificial daylight, CIE Illuminant D75, with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles in lieu of D65. (See 6.6.1). The map case shall also be a good match to the standard sample with a secondary source simulating the spectral quality of incandescent lamplight, CIE Illuminant A, with a color temperature of 2856K (± 200).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense

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Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The map case is used to carry maps, photographs and equipment for sketching.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. The specific issue of individual documents referenced (see 2.2 and 2.3).
c. When first article is required (see 3.1 and 4.2).
d. Color required (see 3.2.2, 3.2.3).
e. Conformance inspection and acceptance quality limits (AQL) (see 4.3).
f. Inspection conditions (see 4.3.1).
g. Packaging requirements (see 5.1).

6.3 First article. When a first article inspection is required (see 3.1), it will be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article

6.4 Standard sample. For access to samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.6 Visual shade matching. In 2017, Option A of AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5°) angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens.

6.6.1 In 2022, the U.S. Military (All Services) accepted D65 as the new preferred light source for visual shade matching due to the supply chain issues with D75. DD1222's should reflect which light source is used. If the light source is not indicated on the DD1222, D65 will be assumed by the Government Shade Laboratory.

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6.7 Known sources for the envelopes as specified in 3.3.7 are:

Polyethylene sheets or tubing: AAA Brand Plastic Industries,
4041 Ridge Avenue, Bldg. 31, Scotts Lane, Philadelphia, PA 19129

Cellulose acetate or cellulose acetate butyrate: Almac Plastics Inc.,
8 Thompson Road, Windsor, CT 06028

6.8 Subject keyword listing.

Carrying
Envelopes
Sketching equipment

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - GL
Navy - NU
Air Force - 11

Preparing activity:
DLA-CT

(Project: 8460-2026-001)

Review activities:
Navy - MC

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